Spatial Interpolation Model

Inverse Distance Weighted Interpolation for Flood Risk Mapping

MKM Research Labs

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Contents

1 Executive Summary

The Spatial Interpolation Model provides geospatial mapping of river gauge observations to individual property locations using Inverse Distance Weighting (IDW). It is a core infrastructure component of the MKM PhysicalRisk platform, enabling property-level flood depth estimation from sparse gauge networks.

The model constructs a KDTree spatial index over gauge coordinates for efficient nearest-neighbour queries, then applies IDW interpolation with a configurable power parameter to produce water surface elevation (WSE) estimates at arbitrary target locations. A haversine distance function ensures correct great-circle distance calculations on the Earth's surface.

The model also supports polyline projection (nearest point on a river centreline) and spatial correlation matrix construction for portfolio-level risk aggregation.

Key risk: IDW assumes smooth spatial variation and does not capture terrain discontinuities (e.g. flood barriers, embankments). Accuracy degrades rapidly with increasing distance from the nearest gauge.

2 Model Purpose and Scope

2.1 Purpose

The Spatial Interpolation Model serves three functions:

- **Gauge-to-property mapping**: Interpolate flood water surface elevation from observed gauge readings to individual property locations.
- **Spatial indexing**: Build efficient KDTree structures for nearest-neighbour search over large gauge networks.
- **Portfolio correlation**: Construct spatial correlation matrices between properties for aggregated risk calculations.

2.2 Scope

The model handles:

- Point-to-point IDW interpolation with configurable power parameter.
- KDTree construction from latitude/longitude gauge data.
- Haversine great-circle distance calculations.
- Nearest point projection onto river polylines.
- Exponential distance-decay correlation matrices.

3 Methodology

3.1 Inverse Distance Weighting

The IDW estimator at a target point $\mathbf{x}_0$ is:

$$\hat{z}(\mathbf{x}_0) = \frac{\sum_{i=1}^{n} w_i \, z_i}{\sum_{i=1}^{n} w_i}, \qquad w_i = \frac{1}{d(\mathbf{x}_0, \mathbf{x}_i)^p} \tag{1}$$

where z_i is the observed value at gauge i, $d(\cdot)$ is the haversine distance, and p is the power parameter (default $p = 2$).

If any gauge lies within a minimum distance threshold (default 1 m), its value is returned directly without interpolation.

3.2 Haversine Distance

Great-circle distance between two points (ϕ_1, λ_1) and (ϕ_2, λ_2):

$$d = 2R \arctan\left(\sqrt{a},\ \sqrt{1-a}\right), \qquad a = \sin^2\frac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2\frac{\Delta\lambda}{2} \tag{2}$$

where $R = 6{,}371{,}000\,\text{m}$ is the Earth's mean radius.

3.3 KDTree Spatial Index

Gauge coordinates are converted to radians and indexed using `scipy.spatial.cKDTree` for $O(\log n)$ nearest-neighbour queries.

3.4 Spatial Correlation

The correlation between properties i and j decays exponentially with distance:

$$\rho_{ij} = \rho_0 \exp\left(-\frac{d_{ij}}{L}\right) \tag{3}$$

where ρ_0 is the base correlation and L is the correlation length scale.

4 Inputs and Outputs

4.1 Inputs

- Gauge locations: latitude, longitude, elevation (from gauge metadata).
- Gauge flood observations: flood depth per gauge for a given event.
- Target property location: latitude, longitude, elevation.
- IDW power parameter p (default 2.0).
- Minimum distance threshold (default 1.0 m).
- Correlation distance and base correlation (for portfolio aggregation).

4.2 Outputs

- Interpolated water surface elevation at target location.
- Nearest gauge distance and identity.
- Spatial correlation matrix (for portfolio risk).

5 Assumptions and Limitations

5.1 Assumptions

1. Flood water surface varies smoothly between gauge locations.
2. The haversine formula provides sufficient distance accuracy at the spatial scales of interest (¡ 100 km).
3. Gauge observations are contemporaneous for a given flood event.
4. The IDW power parameter is constant across the interpolation domain.

5.2 Limitations

1. IDW cannot model terrain barriers (embankments, levees, ridgelines).
2. Interpolation quality degrades with sparse gauge networks.
3. The flat-Earth approximation used in polyline projection introduces errors at distances > 50 km.
4. No anisotropy: the model treats all directions equally, whereas flood propagation follows river valleys.

6 Sensitivity Analysis

This section analyses the sensitivity of interpolated WSE to each key input parameter.

7 Validation Questions

7.1 Q1 — Purpose

"Is the model being used for its intended purpose?"

Yes. The model is used exclusively for spatial interpolation of flood observations from gauge networks to property locations, consistent with its design scope.

7.2 Q2 — Conceptual Soundness

"Is the model conceptually sound?"

IDW is a well-established geostatistical interpolation method. The haversine distance formula is exact for the Earth's geometry. The KDTree provides optimal nearest-neighbour search complexity.

7.3 Q3 — Data Quality

"Is the data quality adequate?"

Gauge metadata (coordinates, elevation) is sourced from EA and SEPA official records. Quality depends on the upstream gauge data pipeline.

7.4 Q4 — Developmental Evidence

"Is there adequate developmental evidence?"

The implementation follows standard scipy spatial algorithms (cKDTree, cdist). Unit tests verify haversine accuracy, IDW convergence, and KDTree correctness.

7.5 Q5 — Outcomes Analysis

"Are outcomes reasonable?"

Interpolated WSE values are validated against held-out gauge readings and known flood event observations.

7.6 Q6 — Ongoing Monitoring

"What ongoing monitoring is required?"

Interpolation error should be tracked against actual flood observations. The power parameter p should be reviewed annually.

7.7 Q7 — Benchmarking

"Has the model been benchmarked?"

IDW results are compared against kriging and nearest-neighbour baselines.

7.8 Q8 — Monitoring

"What type of monitoring is required?"

Quarterly review of interpolation residuals against observed flood depths. Annual review of gauge network density and spatial coverage.

7.9 Q9 — Overall Risk

"Is the overall model risk acceptable?"

The model is a standard geostatistical tool with well-understood limitations. Overall risk is Low, provided gauge density is sufficient.